COAL LAB – FALL 2026
LAB #05 – 23K-0060

TASK – 1:

Code;

```
INCLUDE Irvine32.inc

.data
arrayB BYTE 12h, 24h, 36h, 48h, 5Ah
arrayW WORD 1111h, 2222h, 3333h, 4444h

.code main PROC

    movzx eax, arrayB[0]
    mov edx, OFFSET arrayB
    mov ebx, eax
    mov edx, eax
    mov edx, OFFSET arrayB
    movzx eax, BYTE PTR [edx]
    call WriteHex
    call Crlf

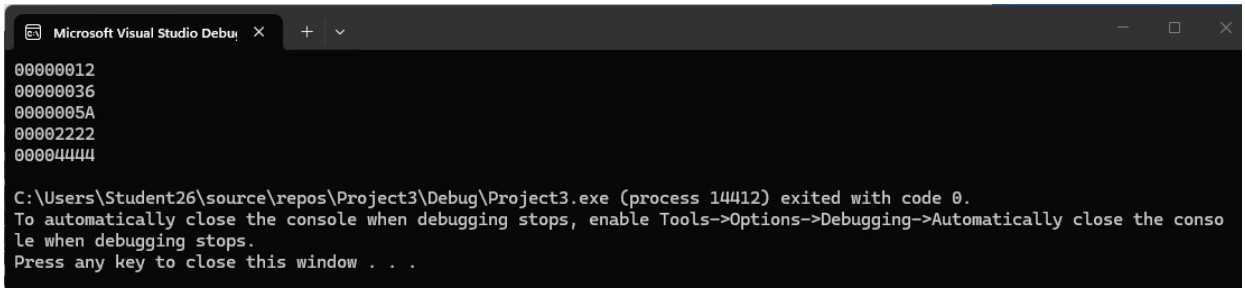
    mov edx, OFFSET arrayB
    movzx eax, BYTE PTR [edx+2]
    call WriteHex
    call Crlf

    mov edx, OFFSET arrayB
    movzx eax, BYTE PTR [edx+4]
    call WriteHex
    call Crlf

    mov edx, OFFSET arrayW
    movzx eax, WORD PTR [edx+2]
    call WriteHex
    call Crlf
```

```
    mov edx, OFFSET arrayW
    movzx eax, WORD PTR [edx+6]
    call WriteHex
    call Crlf
    exit
main ENDP
END main
```

Output;



TASK – 2:
Code;

```
INCLUDE Irvine32.inc

.data numbers BYTE 10, 20, 30, 40, 50, 60
marks WORD 1000h, 2000h, 3000h
values DWORD 11111111h, 22222222h

.code main PROC

    mov esi, OFFSET numbers
    mov edi, OFFSET marks
    mov ebx, OFFSET values

    mov eax, esi
    call WriteHex
    call Crlf

    mov eax, TYPE numbers
    call WriteDec
    call Crlf

    mov eax, LENGTHOF numbers
```

```
    call WriteDec
    call Crlf

    mov eax, SIZEOF numbers
    call WriteDec
    call Crlf

    mov eax, edi
    call WriteHex
    call Crlf

    mov eax, TYPE marks
    call WriteDec
    call Crlf

    mov eax, LENGTHOF marks
    call WriteDec
    call Crlf

    mov eax, SIZEOF marks
    call WriteDec
    call Crlf

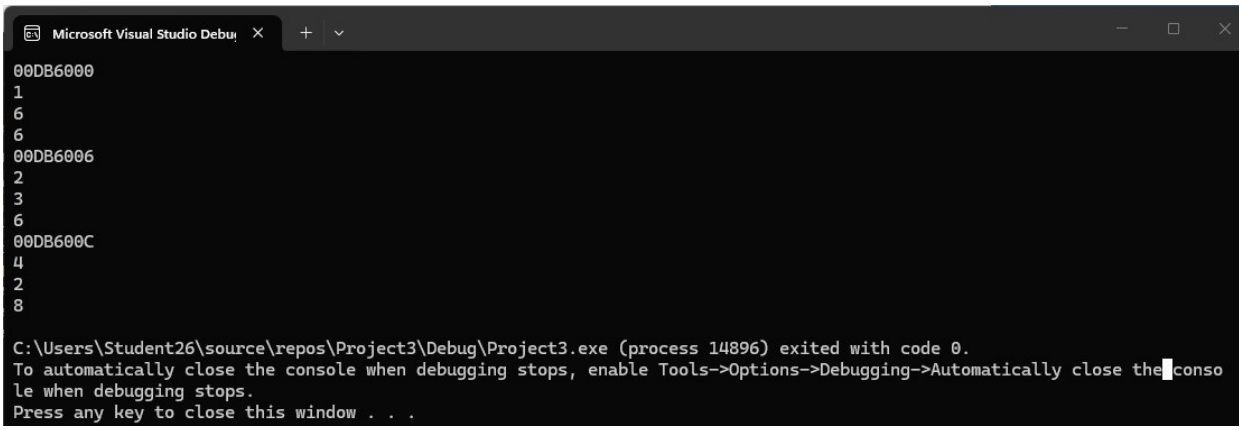
    mov eax, ebx
    call WriteHex
    call Crlf

    mov eax, TYPE values
    call WriteDec
    call Crlf

    mov eax, LENGTHOF values
    call WriteDec
    call Crlf

    mov eax, SIZEOF values
    call WriteDec
    call Crlf
    exit
main ENDP
END main
```

Output;



TASK – 3:
Code;

```
INCLUDE Irvine32.inc

.data
dataVal DWORD 89ABCDEFh
wordList WORD 1357h, 2468h

.code main PROC

   movzx eax, WORD PTR dataVal
   call WriteHex
   call Crlf

   movzx eax, WORD PTR [dataVal+2]
   call WriteHex
   call Crlf

   movzx eax, BYTE PTR dataVal
   call WriteHex
   call Crlf

   mov eax, DWORD PTR wordList
   call WriteHex
   call Crlf
   exit
main ENDP
END main
```

Output;



TASK – 4:
Code;

```
INCLUDE Irvine32.inc

.data
buffer BYTE 15h, 25h, 35h, 45h, 55h

.code main PROC

   mov esi, OFFSET buffer

   mov ecx, LENGTHOF buffer
   mov ebx, esi

L1:
   movzx eax, BYTE PTR [ebx]
   call WriteHex
   call Crlf
   inc ebx
   loop L1

   mov esi, OFFSET buffer

   movzx eax, BYTE PTR [esi]
   mov BYTE PTR [esi+1], 66h
```

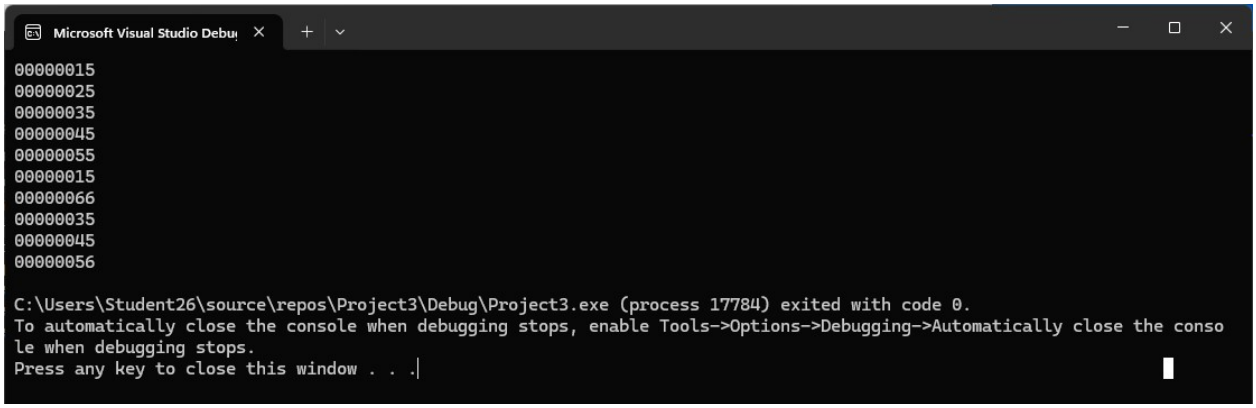
```
    inc BYTE PTR [esi+4]

    mov ecx, LENGTHOF buffer
    mov ebx, esi

L2:
    movzx eax, BYTE PTR [ebx]
    call WriteHex
    call Crlf
    inc ebx
    loop L2
    exit
main ENDP
END main
```

Output;

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TASK – 5:
Code;

```
INCLUDE Irvine32.inc

.data
arrayB BYTE 5, 10, 15, 20, 25
arrayW WORD 100h, 200h, 300h, 400h, 500h
arrayD DWORD 1000h, 2000h, 3000h, 4000h, 5000h

.code main PROC

    mov esi, OFFSET arrayB
```

```
    mov ecx, LENGTHOF arrayB

L1:
    movzx eax, BYTE PTR [esi]
    call WriteDec
    call Crlf
    add esi, TYPE arrayB
    loop L1

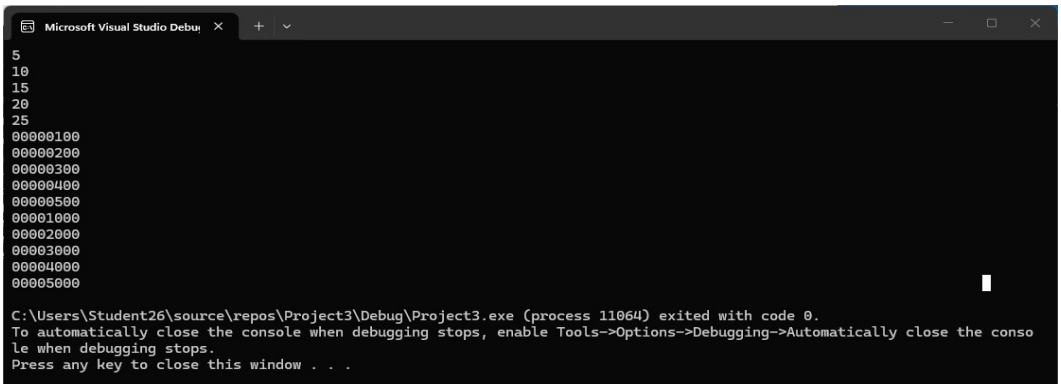
    mov esi, OFFSET arrayW
    mov ecx, LENGTHOF arrayW

L2:
    movzx eax, WORD PTR [esi]
    call WriteHex
    call Crlf
    add esi, TYPE arrayW
    loop L2

    mov esi, OFFSET arrayD
    mov ecx, LENGTHOF arrayD

L3:
    mov eax, DWORD PTR [esi]
    call WriteHex
    call Crlf
    add esi, TYPE arrayD
    loop L3
    exit
main ENDP
END main
```

Output;



TASK – 6:
Code;

```
INCLUDE Irvine32.inc

.data
arrayB BYTE 11, 22, 33, 44, 55, 66
arrayW WORD 1000h, 2000h, 3000h, 4000h, 5000h, 6000h
arrayD DWORD 11111111h, 22222222h, 33333333h, 44444444h

.code main PROC

    mov esi, 0
    movzx eax, arrayB[ESI]
    call WriteDec
    call Crlf

    mov esi, 2
    movzx eax, arrayB[ESI]
    call WriteDec
    call Crlf

    mov esi, 4
    movzx eax, arrayB[ESI]
    call WriteDec
    call Crlf

    mov esi, (1 * TYPE arrayW)
    movzx eax, WORD PTR arrayW[ESI]
    call WriteHex
    call Crlf

    mov esi, (3 * TYPE arrayW)
    movzx eax, WORD PTR arrayW[ESI]
    call WriteHex
    call Crlf

    mov esi, (5 * TYPE arrayW)
    movzx eax, WORD PTR arrayW[ESI]
    call WriteHex
    call Crlf
```

```
    mov esi, (1 * TYPE arrayD)
    mov eax, DWORD PTR arrayD[ESI]
    call WriteHex
    call Crlf

    mov esi, (3 * TYPE arrayD)
    mov eax, DWORD PTR arrayD[ESI]
    call WriteHex
    call Crlf
    exit
main ENDP
END main
```

Output;

